

PHYSIOL 578: HW 1.2

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Preface

You may work in groups but write your answer(s) in your own words with your own figures/calculations where needed. Develop answers using course content, not outside resources.

- Permitted use of genAI
 - Asking for a definition or explanation of a general term
 - Receiving feedback on the clarity of an answer you have already written
 - Grammar, organization, or copy-editing that does not add scientific context
- Not permitted
 - Uploading questions, diagrams or graphical elements and asking a model to answer or interpret it
 - Asking a model to perform calculations, identify sequence domains, analyze traces or generate scientific reasoning
- Process
 - If using genAI, provide as an appendix (a) the original independent answer, (b) the tool/model used, (c) the prompts used and output, and (d) a brief summary of what you changed in response to the genAI results.
 - If you did not use genAI for this assignment, check this box
☐ “I did not use generative AI for this assignment”

Question 1

The amino acid sequence below is for a channel with substantial homology to a classic voltage-gated potassium channel alpha subunit. Major hydrophobic domains are highlighted yellow. A splice variation has inserted a series of negative residues near the mouth of the channel, which are highlighted in blue.

NH2..MTVMHDDHECLLGPNPKRMRYFDNEYFFDRNRPSFDAILYYYQSGRLRRPVNVLDMFSEEIKFYELGEEAM
EKFREDEGFIKEERPLPEKEYQRQVWLLFEYPESSGPARIIVSMVILISIVIFCLETLPELKDDKDFTGTIHRIDNTTVIYT
SNIFTDPFFIVETLCIIWFSFELVVEFFACPSKTDFFKNI MNFIDIVAIIPYFITLGTETAEQEGNQKGEQATSLAILRVIRLVRV
FRIFKLSRSSKGLQILGQTLKASMRELGLLIFFLFIGVILFSSAVYFAAEEAEFIPDAFWWAVVSMTTVGYGDMYETEGEEP
VTIGGKIVGSLCAIAGLTIALPVPVIVSNFNYFYHRQAQLLHVSSPNLASDSDLSRRSSSTISKSEYMEIEEDTQLRTLEYFFD
RNREEKFIPDYYQSGRLRPVNRDMFSEEIKDEGFIETVVINISGLRPEYLLWVQRKEEERDEYQTTVINDRITKESIK..COOH

For each of the features below, identify the highlighted sequence segment most relevant to the feature and provide a 1-sentence rationale for your choice.

(a) Pore

Sequence: WWAVVSMTTVGYYG. The pore is a small hydrophobic domain between S5 and S6 that contains the consensus sequence TVGYG that dictates potassium permeability

(b) Primary voltage sensor

Sequence: ILRVIRLVRVFRIFKLSRSSK. The voltage sensor is a series of positively charged amino acids in S4.

(c) Contains the gating hinge

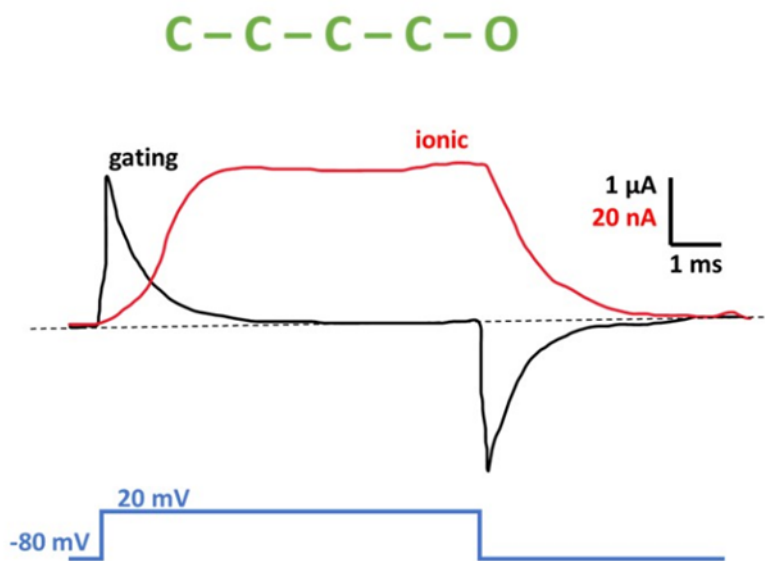
Sequence: GGKIVGSLCAIAGLTIALPVPVIVSNFN. The PxP (PVP in this case) sequence residues form a flexible hinge in S6.

(d) Increases single channel conductance

Sequence: ETEGEE. The splice insert places many negatively charged amino acids near the mouth of the channel (outside S6), increasing conductance.

Question 2

You record from a cell overexpressing a voltage-gated potassium channel that follows the state model shown (green), with voltage-dependent rate constants between each state. Under the whole-cell configuration using normal ionic conditions, you separately record the ionic current (red) and gating current (black, note the change in scale) due to a depolarizing voltage step (blue). Using this figure, briefly answer the questions below (2-3 sentences each).



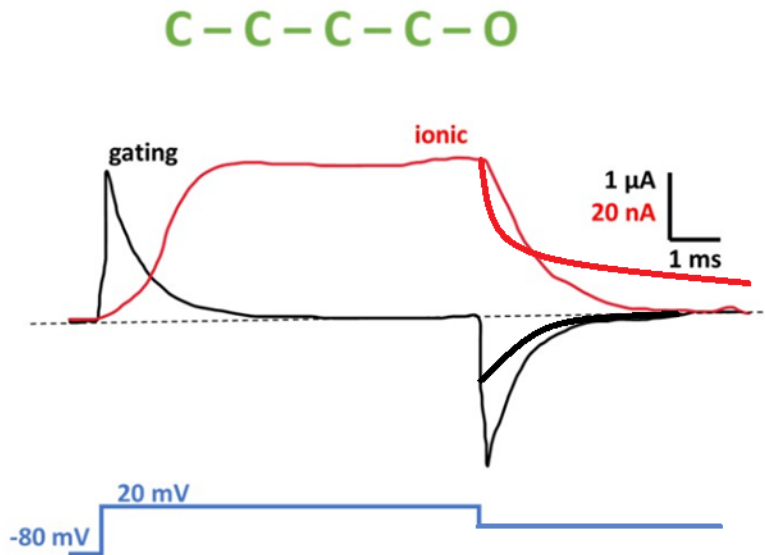
(a) During the activation step, why is gating current transient but ionic current sustained? Why does the gating current appear to peak long before the ionic current?

The gating current reflects the outward movement of the positive charge of the voltage sensor in the channel associated with opening of the activation gate. It is transient because the movement of the sensor is limited. After it stops moving, the current disappears. The gating current comes long before the ionic current because the channel must make 4 conformation changes (from C1 to C2 to C3 to C4 to O) to open in order to let current flow.

(b) At the end of the voltage step, why does the deactivation of the ionic current coincide with the rebound of the gating current (in other words, why is there no delay in this case)?

The deactivation coincides with the gating current because there is only one conformation change from open to close (from O to C4).

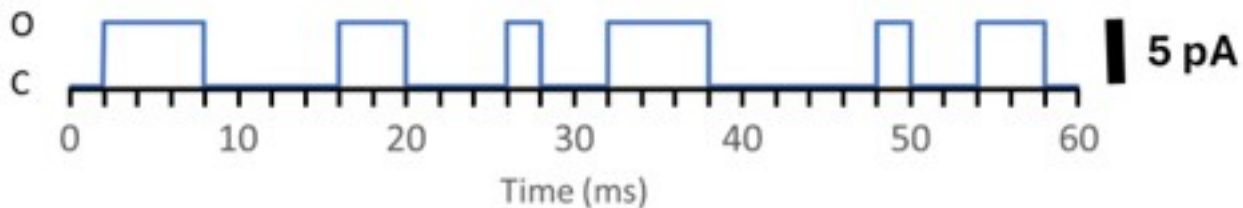
(c) A second voltage protocol returns the membrane from +20mV to -40mV rather than -80mV. Using the state model and principles from class, draw the predicted OFF gating current and ionic deactivation current traces relative to the originals. Explain the sign, approximate relative size, and timing of each change.



The ionic current is still positive since positive K^+ ions are leaving the cell. Conversely, the gating current is negative as the voltage sensor moves inward. The magnitude of both ionic and gating currents are reduced due to the decrease in driving force. Accordingly, the decrease in both current magnitudes are slower. The gating current eventually goes to 0 while ionic current is sustained above zero as there is still some ionic flow.

Question 3

You record the following single channel current trace (idealized) from a typical voltage-gated potassium channel at steady-state in an outside-out patch at +20 mV with equal 150 mM KCl on both sides of the membrane. Scale bar on the right shows the single channel current amplitude. Calculate the following. Include relevant equations and correct units.



(a) Single channel conductance

$$I = \gamma(V_m)$$

$$5\text{pA} = \gamma(20\text{mV})$$

$$\gamma = \frac{5\text{pA}}{20\text{mV}} = \boxed{250\text{pS}}$$

(b) Open probability

$$P_{\text{open}} = \frac{t_{\text{open}}}{t_{\text{total}}}$$

$$= \frac{(6 + 4 + 2 + 6 + 2 + 4)\text{ms}}{60\text{ms}} = \frac{24}{60} = \boxed{0.4}$$

(c) Mean Open Duration and Mean Closed Duration

$$\mu_{\text{open}} = \frac{t_{\text{open}}}{T_{\text{open}}}$$

where T_{open} is number of times the channel was open

$$\mu_{\text{open}} = \frac{24\text{ms}}{6} = \boxed{4\text{ms}}$$

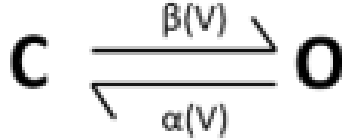
$$\mu_{\text{closed}} = \frac{t_{\text{closed}}}{T_{\text{closed}}}$$

where T_{closed} is number of times the channel was closed

$$\mu_{\text{closed}} = \frac{(60 - 24)\text{ms}}{7} = \boxed{5.14\text{ms}}$$

(d) Assuming an ideal two-state model (Figure 1), estimate α and β from the dwell intervals and then predict open probability from those rates. How does this prediction compare with your calculation in (b), and explain if they are different?

Figure 1: Ideal two-state model.



$$\alpha = \frac{1}{\mu_{\text{open}}} = \frac{1}{4\text{ms}} = \boxed{0.25\text{ms}^{-1}}$$

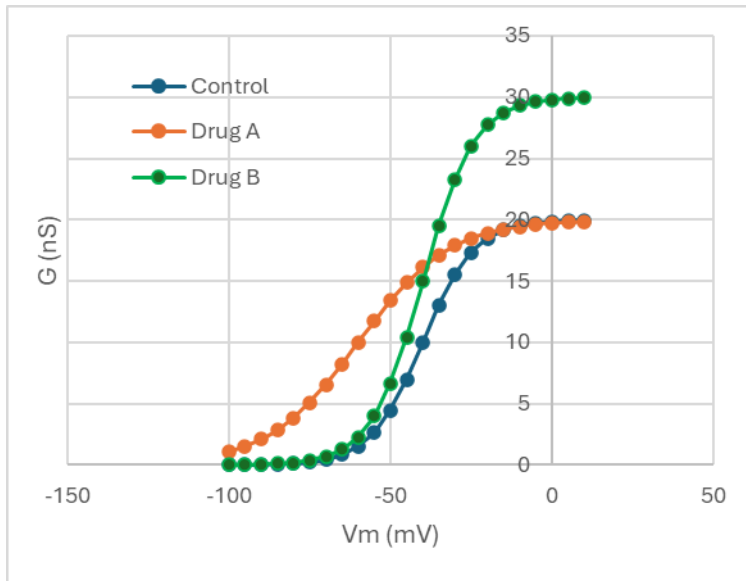
$$\beta = \frac{1}{\mu_{\text{closed}}} = \frac{1}{5.14\text{ms}} = \boxed{0.194\text{ms}^{-1}}$$

$$\hat{P}_{\text{open}} = \frac{\beta}{\alpha + \beta} = \frac{\frac{7}{36}}{\frac{6}{24} + \frac{7}{36}} = \frac{7}{36} = \boxed{0.4375}$$

The estimated open probability using α and β differs from open probability calculated from t_{open} . This is likely due to the stochasticity of the system. If we were to measure the current for longer, we would see open probability using α and β converge to open probability calculated from t_{open} .

Question 4

You record from a cell heterologously overexpressing a single type of voltage-gated potassium channels (i.e., many channels of a single type are expressed in each cell). You record under Whole-Cell conditions with normal physiological intracellular and extracellular solutions (namely, $E_K = -80$) to get the macroscopic steady-state conductance vs voltage curves below under control conditions and in the presence of Drug A and Drug B. Assume the drugs only effect this potassium channel. Answer the following questions.



(a) The resting membrane potential for this cell is -60 mV. How would Drug A change the resting potential (more positive, more negative), if at all?

Drug A would make the resting potential more negative. Because Drug A has higher G at $V = -60\text{mV}$ than Control, more K^+ ions are allowed to leave the cell thus hyperpolarizing the cell.

(b) How would Drug B change the resting potential, if at all?

Drug B would not change the resting potential much. Because Drug B has relatively the same G at $V = -60\text{mV}$ compared to Control, there would be same amount of K^+ ions leaving.

(c) Estimate V_{half} from the curves for control conditions and when exposed to Drug A and Drug B.

$V_{\text{half,C}}$ for control condition is V at which $G = 10\text{nS}$.

$$V_{\text{half,C}} \approx -40\text{mV}$$

$V_{\text{half,A}}$ for when exposed to drug A is also V at which $G = 10\text{nS}$.

$$V_{\text{half,A}} \approx -65\text{mV}$$

$V_{\text{half,B}}$ for when exposed to drug B is V at which $G = 15\text{nS}$.

$$V_{\text{half,B}} \approx -40\text{mV}$$

(d) In addition to V_{half} , what parameter in the Boltzmann equation does Drug A appear to impact? Provide at least one hypothesis for how Drug A could affect the voltage sensor S4 to achieve this effect.

Drug A seems to impact the "voltage sensitivity" parameter q in the Boltzmann equation. Perhaps Drug A is binding to the voltage sensor S4 in a way that makes it less polar.

(e) During lab meeting, some argue that Drug B increases channel number, others say it increases single-channel conductance, still others say it increases maximum open probability. State whether the present G-V data can distinguish between these three possibilities. Briefly, propose one additional experiment that would help discriminate among the viable explanations.

The G-V plot alone cannot determine the cause of the change in channel conductance of a whole-cell. This is due to the fact that $G = N * \gamma * P$ and thus is a single output of 3 different variables. We can discriminate among the explanations by doing

a voltage clamp on a single channel to get γ and P .